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Developing Marginal Fields in the Empty Quarter

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Abstract

An Operating Company has a portfolio of small and widely dispersed oil reservoirs in the remote desert areas southeast of Abu Dhabi at the edge of the Empty Quarter. Marginal fields are characterized by small reserves, shorter production plateaus and lack of nearby infrastructure. Conventional project execution strategies, technologies, and field architecture adopted for developing typical large Middle East oil fields, do not necessarily apply to marginal fields.

Adopting a "Develop while Appraise" approach and "Fit for Purpose" project execution strategy reduced the time from discovery to first oil. Proven technologies were used rather than piloted emerging technologies, to develop marginal fields. Standardizing engineering design of well site facilities, use of bulk procurement of tie-in materials and Call-off type contracts, resulted in a shorter project schedule. This enhanced marginal field project economics and allowed an E&P Company with a lean agile organization to develop previously untapped small oil reserves in an efficient way.

This paper addresses the key strategies implemented in developing these marginal fields. A case study discusses a marginal oil field that was developed from "Discovery to First Oil" in just over two years.

Introduction

The Empty Quarter ([Rub Al-Khali](#) in Arabic) extends across the southern part of the Arabian Peninsula and crosses borders into four countries: Saudi Arabia, Yemen, Oman, and the United Arab Emirates. The map of the Empty Quarter is shown in [Fig.1](#) (Encyclopedia Britannica). It contains the largest continuous desert in the world. Because of the dry conditions and extremely high temperatures, the Empty Quarter is uninhabited and largely unexplored (Encyclopedia Britannica).



Figure 1—The Empty Quarter

The Empty Quarter has significant hydrocarbon reserves. The largest conventional oil field in the world, Al-Ghawar field of Saudi Arabia lies along the northeastern edge of the Empty Quarter. At the other end of the scale, in terms of field size, are the marginal fields discussed in this paper.

Al Dhafra Petroleum Company, which was established in 2013, is assigned a concession area southeast of Abu Dhabi at the eastern edge of the Empty Quarter. The primary focus of the Company is to explore and develop the marginal fields in the concession area. Fig. 2 shows the concession area. Some of the fields in the concession area were discovered and tested in the 1990s.



Figure 2—The Company Concession Area

The definition of what is considered a marginal field differs between the various oil producing regions of the world and even from one company to another. In general, marginal fields are characterized by:

- Small size reserves.
- Short production plateaus.
- Lack of infrastructure in the vicinity.
- High development cost and poor economics, e.g. the Unit Technical Cost (UTC) will exceed the Company threshold.

Conventional project execution strategies, technologies, and field architecture adopted for developing typical large Middle Eastern oil fields, do not necessarily apply to marginal fields.

Surface Facilities Hub

The first development target for the Company was the field which was estimated to have the largest Original Oil in Place (OOIP). The producing formations are carbonate reservoirs of the Lower Cretaceous Group formation in the Liwa basin area. The field is referred to as Field A in Fig. 3. During the Assess/Select stages of the project, studies were conducted to evaluate options for the export of Field A crude oil. Several options were evaluated. These include:

- Full stabilization of the crude oil at Field A.
- Partial stabilization of the crude oil at Field A.
- Exporting production without stabilization using Multiphase Pumps (MPPs) to the facilities of a nearby sister company for separation and stabilization.

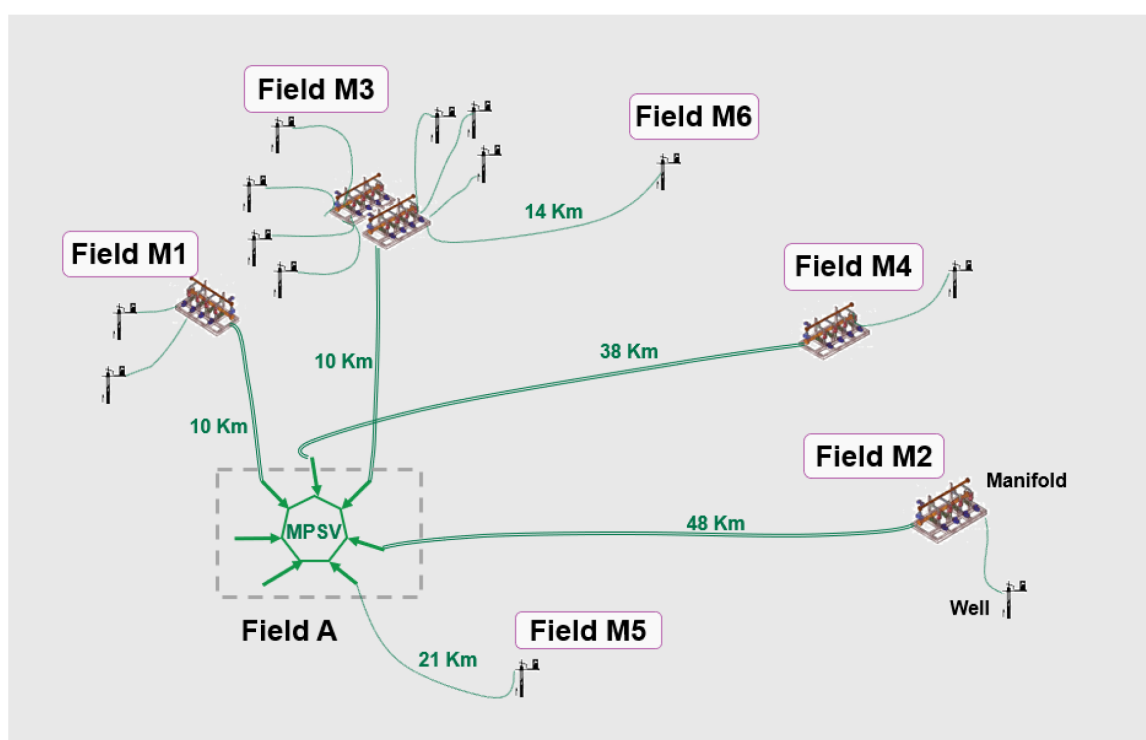


Figure 3—Field A is the Hub for Marginal Fields in the Concession Area

The last option was selected as the optimum option and most cost-effective solution.

Individual flowlines of the producing wells from the main field are gathered at the Central Gathering Facilities (CGF), through Multiport Selector Valves (MPSVs). Each MPSV can accommodate up to a maximum of seven flowlines. Production testing of individual wells is carried out by Multiphase Flow Meters (MPFMs) provided at the CGF. Typically, each oil well is tested once a month for a duration of 24 hours. A maximum of two wells can be tested at a time. Therefore, three MPFMs (two in service and one standby) are provided at the CGF. Testing of wells is pre-programmed through the process control system and can be monitored and recorded from the control room.

The CGF was designed to be a hub to support and receive production from the small marginal fields in the southeast area of Abu Dhabi Emirate. The CGF and infrastructure were executed using an EPC type contract. The flowlines between the wells and the CGF were not part of the EPC Contract but were executed using a Contract on a Call-off Order basis.

The multiphase flowlines from the wells to the CGF are designed in accordance with ASME Piping Code B31.8, Class 2. The pipeline material is carbon steel, typically API 5L Grade X65. The flowlines are laid on the surface with no external coating. Three-layer polypropylene coating and sacrificial anodes are provided only at the buried sections of the flowlines when crossing a track or a rig road. In the Sabkha (flat salt), flowlines are supported by pipe hurdles.

To streamline flowlines corridors, the flowlines from various wells are divided into seven well groups (WGs) as shown in Fig. 4.

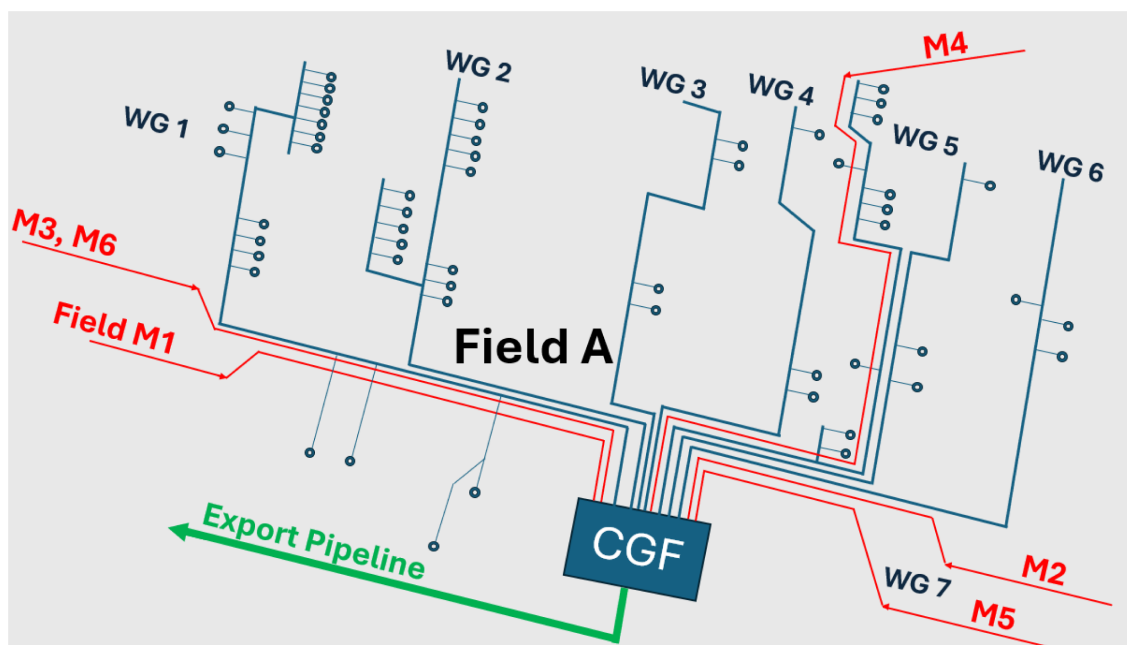


Figure 4—The Gathering System

Production from the main field is transferred using MPPs through an approximately 65 km, 16 in. pipeline to the facilities of a nearby sister company for treatment, processing and export.

Reservoir pressure is maintained by water injection. The water is sourced from an aquifer. Water supply wells are in the vicinity of the CGF. Electrical Submersible Pumps (ESPs) in the water supply wells lift the aquifer water. Water injection surface pumps at the CGF distribute the aquifer water to the water injection wells.

Like the oil flowlines, the water supply lines as well as the water injection lines are laid on the surface with no external coating.

There is no power generation at the CGF. Power is supplied through a 132 kV double circuit overhead line from the country electrical network. Two containerized gas-insulated switchgear bays with 132 kV circuit breaker panels and two 132/33 kV power transformers are utilized to feed the CGF substation. Power supply from the CGF substation to the oil producing wells in the main field is provided by two separate 33kV overhead lines.

Considering the short production plateaus of the marginal fields and the long distances from the CGF, it was more economical to supply power to marginal fields wells by individually dedicated Diesel Generator sets. Power is supplied to the ESPs from the remote marginal fields wells by Diesel Generator sets.

Field A was commissioned in 2019. This was followed shortly by commissioning marginal Field M1.

Discovery to First Oil in Two Years

The remotest marginal field from the CGF was discovered in 2020. The field is designated as Field M2 in [Fig. 3](#). It is located 48 km west of the CGF near the UAE borders with Oman. The first well was tested and proved oil during July 2022.

The surface facilities project was implemented on a fast-track basis. Because of the lead time involved with the procurement of material, the project team had to adopt a different approach ([Nair et al. 2021](#)). The supply of material was investigated by searching for available material in other business units of the group companies. Suitable materials were identified and procured from these business units. In some cases, the procured material did not exactly match with the requirements. For example, pipelines sourced from the other business units had higher wall thickness than what is specified in the design. A 6 in., 50 km long surface-laid pipeline was constructed through the hilly terrains, heavy sand dunes and Sabkhas to flow Field M1 production to the CGF.

The first oil was produced in November 2022. Achieving the first oil in such a relatively short time after discovery is noteworthy. Especially since the Company follows the Group Company's comprehensive approval process as well as the rigorous procurement and contracting procedures. The Company must also comply with the requirements of multiple Shareholders.

Several factors contributed to the successful production of Field M1 including:

- Early engagement of the Shareholders by the Company Field Development Department.
- Reducing the Conceptual Field Development Plan (FDP) and reviewing milestones with the Stakeholders from the standard six milestones to two.
- Approval of engineering documents within two days.
- Early start of flowline construction activities.
- Reducing construction time by working weekends and extending working hours.
- SIMOPS of wellsite construction activities while conducting well "Flow to Clean".
- Effective integrated multi-discipline approach to the development using the limited resources of a lean organization.

Custody Transfer Metering Facilities

To resolve the metering uncertainties associated with MPFMs, the Company recently installed custody transfer metering facilities at the CGF ([Saadawi, H. and Al Mheiri, F. 2024](#)). The project was executed using Design-Build-Operate (DBO) contracting strategy ([Saadawi et al. 2024](#)). The custody metering facilities are owned and operated by the DBO Contractor.

The facilities include a 3-phase Separator, Oil Booster Pumps, a Custody Meter Skid and associated instrumentation. All three phases are separated and metered separately and again mixed at the MPP suction

for further transfer and export to the host facilities. Separated oil in the 3-phase Separator is measured through a Custody Meter Skid as shown in Fig. 5. The Custody Meter Skid's accuracy and uncertainty meet the fiscal transfer requirements.

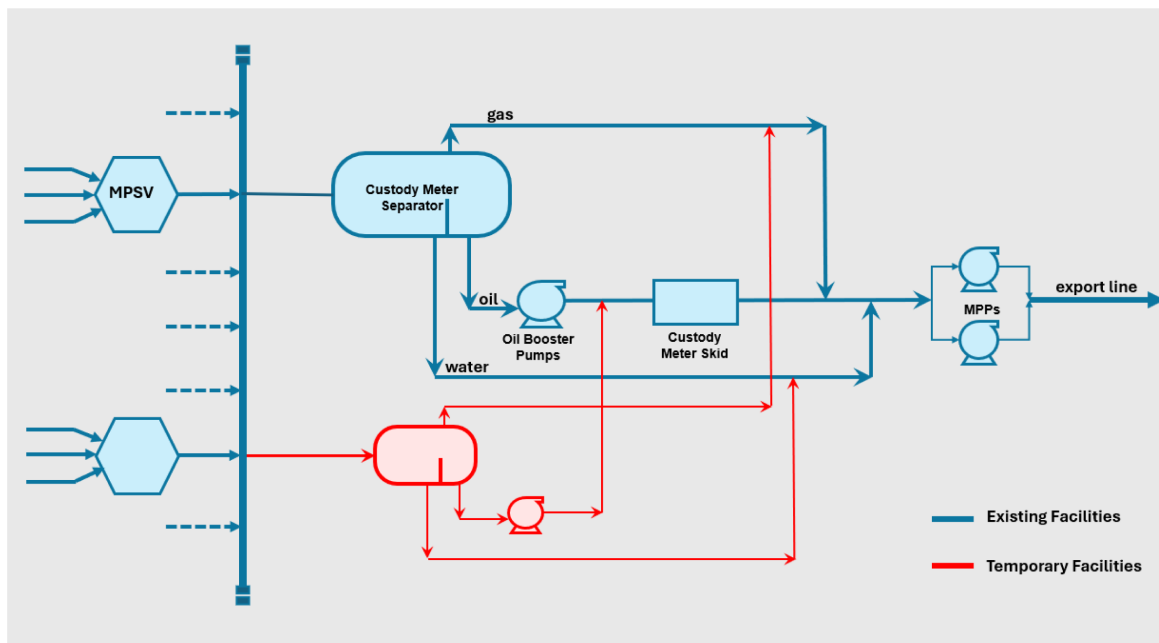


Figure 5—Existing & Temporary Produced Water Separation Facilities

The DBO strategy for executing oil fields projects relies on a safe "Fit for Purpose" approach to design (Saadawi et al. 2024). The use of unnecessary safety factors is avoided. Facilities are designed to industry standards such as API, ASME, ISO and not Company's standards. DBO facilities are typically modular construction, which facilitate mobilization and demobilization at the end of the project life. The project execution schedule is shorter with a DBO contracting strategy than using conventional Front End Engineering Design (FEED) / Engineering, Procurement, and Construction (EPC) approach. This accelerates the date of the first oil and thus improves the economics of marginal fields development. Indeed, DBO approach is suitable for marginal fields with short production plateaus. Furthermore, executing projects using a DBO contracting strategy suits smaller operating companies with lean organizations and limited resources.

Debottlenecking the Existing Central Gathering Facilities

The Company plans to increase crude oil production. With the increase of water cut, the total produced fluids will exceed the liquid handling capacity of the CGF. Plans are underway for three main projects to debottleneck the CGF.

- Temporary Produced Water Separation and Disposal Facilities.
- Produced Water Treatment Facilities.
- Additional MPP and Water Injection Surface Pumps.

These projects are briefly discussed below.

Temporary Produced Water Separation and Disposal Facilities

As an interim measure, a temporary separator will be installed in parallel to the existing Custody Meter Separator. Furthermore, an Oil Booster Pump will pump the separated oil upstream the Custody Meter Skid.

This is shown in Fig. 5. The purpose of this project is to bridge the produced water handling gap while the long-term produced water treatment facilities are being tendered and constructed.

This project will be executed as a variation order to the existing Custody Transfer Metering Facilities DBO Contract described in the previous section. The temporary facilities will operate for 18 months till the final facilities, described in section (b), are commissioned.

Produced Water Treatment Facilities

The objective of this project is to debottleneck the existing facilities by installing produced water separation, treatment & disposal / reinjection facilities at the CGF.

To minimize pre-investment and reservoirs uncertainties, the project scope will be executed in two phases:

- Phase I: Train 1 to cater for the forecast production profile for the next 4 years.
- Phase II: Train 2 to cater for future production profiles (from year 2030 and beyond).

Project scope includes Bulk Separator, Oil Booster Pumps, Electrostatic Coalescer, Custody Meter Skid, Produced Water Treatment Facilities (Hydro-cyclones, Degasser, Vapor Recovery Compressor) and Produced Water Disposal / Re-injection Facilities.

Produced water re-injection will start for selected wells and operate initially for six months. The objective of the stepwise application of produced water re-injection is to access the injection performance before full field deployment.

The project execution strategy is competitive DBO tendering.

Additional MPP and Water Injection Surface Pumps

The project scope includes a third MPP identical to the existing two MPPs and four additional water injection surface pumps, which are also identical to the existing ones.

Because the scope is well defined, the project is being executed using competitive EPC tendering.

Proven Technologies

Emerging piloted technologies can offer significant advantages but can also potentially carry higher risks. Because of the tight economics associated in developing marginal fields, reliable, proven and established technologies were adopted. This is to ensure reliability and simplicity in operation with lower risk.

The table below outlines some of the proven technologies adopted.

Criteria	Traditional Approach	Proven Technology
Production Facilities	Separation Trains, Tanks, Pumps and Compressors	MPPs
Production Manifolds	Conventional Manifolds with Production and Test Headers & Valves	MPSVs
Well Testing	Test Separators	MPFMs
ESP Motors	Induction Motors	Magnetic Motors
Pigging	Barrel-Type Pig Launchers and Receivers	Pigging Valves
Material of Disposal Water Flowlines	Carbon Steel with Corrosion Inhibitors or Internally Coated	Non-metallic (Reinforced Thermoplastics, Glass Reinforced Epoxy)

Criteria	Traditional Approach	Proven Technology
Data Transmission	Fiberoptics	Global System for Mobile Communication (GSM) Wireless Network for remote wells

Using Multiphase Twin-Screw Pumps for transporting 3-phase fluids eliminated the need for separators and full stabilization and processing facilities. This approach leverages existing nearby company infrastructure as host facilities, minimizes upfront capital investment and improves project economics.

MPSVs were used instead of conventional production manifolds. A single MPSV can accommodate up to seven flowlines through it. MPSVs offer compact design by combining the production and test headers in one unit. MPSVs reduce the number of valves, as well as simplify control and operation requirements.

For well testing, MPFMs were used instead of the test separators. Not only did this reduce the footprint and associated piping and instrumentation, but it allowed the operator to program the wells testing plan in the Digital Control System (DCS) and conduct the production tests remotely.

Pigging Valves were used instead of traditional Pig Launchers and Receivers because they have significantly less footprint and reduce the space required for pigging facilities. They also reduced the requirements for infrastructure.

Recently, the Company used a Permanent Magnet Motor (PMM) instead of the conventional induction motor to drive the ESP. The use of a PMM instead of an induction motor can save 15% energy. In some applications, a single PMM can replace triple tandem induction motors; thus, improving the reliability of the ESP installations (Vargas et al. 2024). This is because PMM is more energy intensive and is smaller in size and lighter than an induction motor of the same power rating.

Within the main oil field, Field A, fiber optics were used for data transmission to enable the automation of the wells of Field A. The remote marginal fields are operated unmanned. However, for production monitoring, the GSM wireless network of the local telecommunication company was used for data transmission. This was more cost effective than laying fiberoptics cables in over long distances in the sand dunes. The architecture of the data flow for the remote fields is illustrated in Fig. 6.

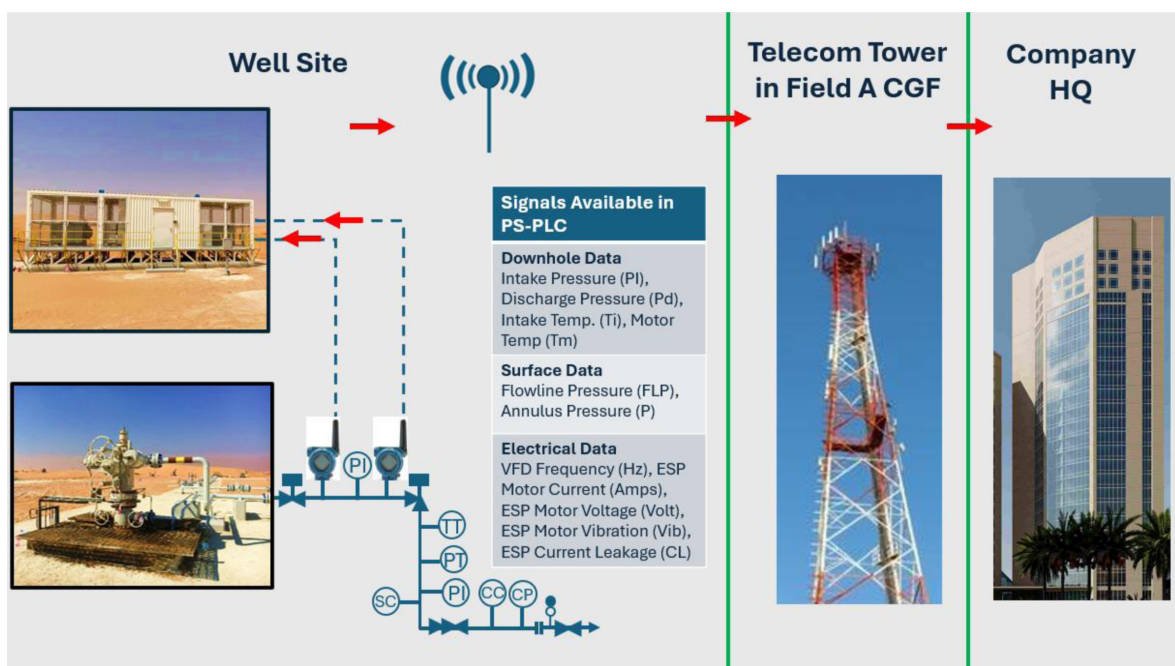


Figure 6—Remote Well Monitoring System Architecture

Fit for Purpose Project Execution Strategy

Selecting the right project execution and contracting strategy is one of the most important decisions in project management. This is even more important in case of marginal fields where project economics are tight. Developing the appropriate contracting strategy depends on several factors such as:

- Project scope and schedule.
- Specific project drivers.
- Market conditions and availability of suitable Contractors.
- Company's own resources.

The different contracting strategies adopted by the Company and the rationale for strategy selection are shown in the table below:

Project Description	Contract Type	Justifications
CGF	EPC – Lump Sum	Scope included infrastructure to develop the CGF as a hub for development of marginal fields
Wells Tie-ins and Flowlines	Call-off Contract	Wells tie-ins are implemented in a phased manner in accordance with the FDP
Custody Transfer Metering Facilities	DBO	• Lean Company organization • Short execution schedule • More cost effective than conventional EPC contract
Supply of Power Skids to Remote Wells	Call-off Contract	Power Skids are supplied as per tie-ins requirements of remote wells
Temporary Produced Water Separation Facilities	Variation to the Existing DBO Contract	• Temporary facilities • Facilities are installed within the plot of the existing Custody Metering Facilities • Avoid introducing another Contractor to operate the facilities
Additional MPP and Water Injection Surface Pumps	EPC – Lump Sum	Scope is well defined; • Additional identical MPP • Additional identical WISPs
Produced Water Treatment Facilities	DBO	The same rationale as for Custody Transfer Metering Facilities

Lean Agile Organization

The Company operating model is based on a lean organization with relatively small core staff. An operating company with a lean organization has fewer layers of management than large operating companies. This streamlines the decision-making process. It can be argued that from a project management perspective, the less the stakeholders, the more efficient project execution is.

The Company is organized such that Operations and Projects are in one department, which ensures that the end user is fully engaged in a new project from the beginning. Projects are executed using an integrated project management team made up of Company and Project Management Consultants (PMC) staff.

The Field Development Department as well as the Operation and Projects Department are under one Senior Vice President. This enables an integrated approach to field development, which is particularly important for marginal fields, where economics are challenging and the team is continuously optimizing the development plans and the design.

The field operation is a lean organization with minimum manning level. Considering this lean organization, the infrastructure of Field A and the CGF were designed as a "Smart Field". The Operator can start and shut down the wells from the control room. He can schedule, plan and conduct well production testing from the keyboard of the operator console.

The Company outsources maintenance and support services. A Maintenance Management Agreement was signed with a Service Provider. The Company also entered into Service Agreements with Call-off Orders with the Original Equipment Manufacturers (OEMs) of the major equipment including the including the suppliers of the MPPs, Integrated Control & Safety System (ICSS) and MFPMs.

The lean and agile organization offers unique opportunities for the fresh graduates joining the Company to accelerate their development. This is because of increased feasibility and more personalized mentorship compared with a larger operating company. Early in their careers, the young graduates are exposed to practical hands-on experience and multitasking.

Conclusions

This paper addresses the key strategies implemented by an E&P Company in developing marginal fields in the remote desert areas southeast Abu Dhabi.

Marginal fields are characterized by small reserves, shorter production plateaus and lack of nearby infrastructure. The Company had to adopt different contracting strategies and suitable development options with the appropriate technologies to develop these marginal fields.

The following factors contributed to improving the economics of marginal field development and ensured projects success:

- Using twin-screw multiphase pumps to transport the 3-phase production has eliminated the need for building full stabilization and processing facilities at the development hub.
- Utilizing the existing facilities of a nearby sister company as host facilities allowed the Company to develop previously untapped oil reserves in an efficient way.
- Having remote operation facilities allows a lean operation organization with minimum manning level.
- The duration between "Discovery" and "First Oil" was reduced by adopting the concept of "Development while Appraise". This proved to be a cost-effective solution to tie-in of marginal fields to the facilities of the main oil field.
- The Company adopted "Fit for Purpose" project execution strategy, with the contracting model selected to suit the type of project.
- Standardizing engineering design of well site facilities, use of bulk procurement of tie-in materials and Call-off type contracts, resulted in shorter project schedule.
- Proven technologies were adopted rather than piloted emerging technologies to develop the marginal fields.
- The lean and agile organization enabled the Company to adjust and adopt different project execution strategies with a "Fit for Purpose" facilities required for marginal fields development. The Company has been able to develop marginal fields with OOIP less than 2 million stock-tank barrels (STB).

Finally, the challenges of marginal fields and the lean organization offer the fresh graduates opportunities for diverse experience and exposure to multitasking early in their career, which accelerates their development.

Acknowledgement

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Nomenclature

CGF	Central Gathering Facilities
DBO	Design-Build-Operate
EPC	Engineering, Procurement and Construction
ESP	Electric Submersible Pump
FDP	Field Development Plan
FEED	Front End Engineering Design
MPFM	Multiphase Flow Meter
MPSV	Multi-port Selector Valve
MPP	Multiphase Pump
OEM	Original Equipment Manufacturer
OOIP	Original Oil in Place
PMM	Permanent Magnet Motor
SIMOPS	Simultaneous Operations
STB	Stock-Tank Barrel
WG	Well Group
WISP	Water Injection Surface Pump

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